Two-Stage Gauge-Equivariant Flows for SU(3) Inverse Blocking

Discussion note for collaborators

27 October 2026

Abstract

We propose a *two-stage* inverse-blocking construction for pure SU(3) lattice gauge theory, using two independently trained conditional, gauge-equivariant normalizing flows. Flow 1 generates smoothed fine links V from coarse links W and added random detail variables; Flow 2 generates original fine links U conditioned on V. The forward transformations $U \mapsto V \mapsto W$ are assumed to have been selected in advance; optimizing the blocking kernel is outside the scope of this note. Both flows are trained from paired configurations obtained by applying the appropriate forward block to native ensembles, and their training problems are separate. We review gauge-equivariant flow elements and the multiscale construction of Abbott *et al.*, contrast them with the prolongation/rethermalization approach of Endres and collaborators, and outline a practical numerical program. The proposed algorithm is not yet a demonstrated numerical result.

1 Goal: two separately trained inverse steps

Let $U_\mu(x) \in \mathrm{SU}(3)$ be the original fine gauge links, distributed according to a target Wilson action $S_{\beta_f}[U]$. We take as given two gauge-covariant forward maps,

$$U \xrightarrow{\mathcal{B}_2} V \xrightarrow{\mathcal{B}_1} W. \tag{1}$$

Here V is the smoothed field at the *same lattice resolution* as U, and W is obtained from V by blocking to a lattice with spacing twice as large. In the current construction,

$$(\mathcal{B}_1[V])_\mu(X) = V_\mu(2X)V_\mu(2X+\hat{\mu}) = W_\mu(X). \tag{2}$$

We do not need to specify or retune the smoothing/blocking transformation $\mathcal{B}_2$ here: its definition is an input to the training and evaluation pipeline.

The proposed inverse map is the composition

$$\boxed{W \xrightarrow[z_1]{F_{1,\theta_1}} V \xrightarrow[z_2]{F_{2,\theta_2}} U.} \tag{3}$$

The stochastic variables z_1 and z_2 are independent draws from tractable reference distributions, while each flow is conditioned on the field supplied by the preceding stage. Training F_1 and training F_2 are separate tasks; one can develop, test, or replace one stage without retraining the other in the first instance.

Flow 1 has a *hard* product-blocking requirement $\mathcal{B}_1[V] = W$ for every generated sample. Flow 2 aims to recover the original fine-field conditional distribution $p(U \mid V)$ associated with the chosen forward map $\mathcal{B}_2$. Whether the second relation $\mathcal{B}_2[U] = V$ is imposed as a hard architectural

constraint or achieved approximately in a learned conditional model is a separate implementation choice that must be tested explicitly. These two distinct notions of constraint should not be conflated.

The intended application begins with coarse links W generated at a suitably RG-matched Wilson coupling β_c. Its coarse-field distribution need not be identical to the marginal obtained by blocking an ensemble at β_f; this is a distinct matching question. The scientific aim is to preserve the long-distance information in the input while generating realistic short-distance fluctuations, with any residual rethermalization measured rather than assumed.

2 Gauge-equivariant normalizing flows

Local gauge transformations act on a fine link as

$$U_\mu(x) \longmapsto g(x)U_\mu(x)g^\dagger(x+\hat{\mu}), \qquad g(x) \in \mathrm{SU}(3). \tag{4}$$

A conditional generative model must respect this transformation law, including the gauge freedom associated with fine sites that are absent from the coarse lattice. A normalizing flow is an invertible transformation of reference random variables with a computable Jacobian (relative to the relevant Haar measure), allowing density evaluation and maximum-likelihood training. Gauge covariance by itself does not guarantee invertibility or a tractable Jacobian.

One useful class of gauge-equivariant link layers builds an oriented loop from an active link L and a gauge-transporting staple S. The quantity LS transforms by conjugation at its basepoint. Schematically,

$$P = \mathrm{Proj}_{\mathrm{SU}(3)}(LS), \qquad P' = f_\theta(P; \text{invariant context}), \qquad L' = P'P^\dagger L. \tag{5}$$

When $f_\theta(hPh^\dagger) = hf_\theta(P)h^\dagger$, the output L' transforms as a link. Concrete spectral and related $\mathrm{SU}(N)$ coupling layers are discussed in Refs. [1, 2]. Equation (5) is an illustrative building block, not a fully specified architecture for either of our conditional models. The conditional context must also be assembled gauge covariantly.

3 Relation to Abbott *et al.* and Endres *et al.*

3.1 Multiscale gauge flows: Abbott *et al.*

Abbott *et al.* [3] construct multiscale gauge-field generators that split coarse links and sample additional fine links using gauge-equivariant conditional flows. The basic split for a coarse link W is

$$V_A = WH^\dagger, \qquad V_B = H, \qquad V_AV_B = W, \tag{6}$$

where H may be drawn from Haar measure or modeled conditionally. New links that do not lie along coarse-link paths are introduced through further conditional steps. The published construction also considers subsequent fine-lattice flow transformations; unless these are constrained, they need not preserve the original product-blocked links.

Our *Flow 1* adopts the key geometric fact in Eq. (6), but is trained specifically on paired (W, V) configurations generated by the *chosen* forward blocking map. Our *Flow 2* is an additional, independent conditional inverse step, returning from smoothed links V to original links U. It should not be absorbed into Flow 1 or replaced by an assumed inverse of the smoothing operation.

3.2 Multiscale equilibration: Endres and collaborators

Endres and collaborators [4, 5] begin with RG-matched coarse configurations, prolongate them through a deterministic gauge-field interpolation (including staged local action minimization), and then equilibrate with a fine-lattice Markov chain. The primary objective is to make the initial fine configurations weakly coupled to slowly relaxing modes; exact sampling of a learned conditional UV distribution is not needed. Their scaling analysis motivates investigating the rethermalization cost of any prolongation scheme, but it does not establish that rethermalization will be unnecessary for our construction.

	Endres multiscale equilibration	Proposed two-stage conditional flow
Coarse starting point	Matched coarse gauge ensemble	Matched coarse ensemble W
First refinement	Deterministic interpolation/minimization	Flow 1: sample $V \mid W$, preserve $\mathcal{B}_1[V] = W$
Recovery of original links	Fine-action equilibration	Flow 2: sample $U \mid V$, trained independently
Fine-action role	MCMC equilibration is integral	MCMC may be used to correct or test resulting U
Key test	Rethermalization of prolongated field	Fidelity of both conditional stages and their composition

4 Flow 1: from blocked links W to smoothed links V

4.1 An exact parameterization of the product constraint

For each coarse link from $2X$ to $2X + 2\hat{\mu}$, introduce a detail link $H_\mu(X)$ on the second half of the path. Define

$$V_\mu(2X + \hat{\mu}) = H_\mu(X), \tag{7}$$

$$V_\mu(2X) = W_\mu(X)H_\mu^\dagger(X). \tag{8}$$

Consequently Eq. (2) holds identically, independently of the stochastic distribution of H. All other fine links are unconstrained detail variables. Collect H and these other links into the aggregate detail variable D, and write

$$D = T_{1,\theta_1}(z_1; W), \qquad V = \mathcal{I}_1(W, D), \qquad \mathcal{B}_1[\mathcal{I}_1(W, D)] = W. \tag{9}$$

This formulation places the normalizing flow entirely on the detail coordinates. A generic fine-link flow applied *after* assembly would break the hard constraint unless each layer were constrained to stay within a fixed-W fiber.

On a periodic L^4 coarse lattice there are $4L^4$ coarse links and $64L^4$ fine links on the $(2L)^4$ lattice. Thus D consists of $60L^4$ SU(3) variables before quotienting by gauge transformations: $4L^4$ split-link variables and $56L^4$ other fine links.

Under independent fine-site gauge transformations, $H_\mu(X)$ transforms between $2X + \hat{\mu}$ and $2X + 2\hat{\mu}$, whereas $W_\mu(X)$ transforms between $2X$ and $2X + 2\hat{\mu}$. Equation (7) therefore respects gauge covariance. A viable conditional flow must also handle the new fine-site gauge freedoms; Haar reference variables and link/staple-based equivariant layers provide a natural starting point.

4.2 Independent supervised training for Flow 1

Generate ordinary fine Wilson configurations and apply the chosen forward maps:

$$U \sim p_{\beta_f}(U), \qquad V = \mathcal{B}_2[U], \qquad W = \mathcal{B}_1[V]. \tag{10}$$

The paired data (W, V) determine the training target for Flow 1. With $D = D(W, V)$ recovered from the constraint-preserving decomposition, a conditional maximum-likelihood objective is

$$\mathcal{L}_1(\theta_1) = -\mathbb{E}_{(W,V)}\big[\log q_{1,\theta_1}(D(W,V) \mid W)\big]. \tag{11}$$

Densities are taken with respect to products of Haar measures on the detail links. The target is the distribution of *smoothed* fine configurations conditioned on blocked configurations, not the fine Wilson action evaluated directly on V. Model validation should include both gauge-invariant observables of V and checks that the blocked W is reproduced exactly.

4.3 Flow 1 architecture: hierarchical constrained refinement

The multiscale refinement of Abbott *et al.* [3] is a natural starting point. Rather than predict all $60L^4$ detail links at once, introduce them through four successive doublings of the lattice extents,

$$L^4 \longrightarrow (2L)\, L^3 \longrightarrow (2L)^2 L^2 \longrightarrow (2L)^3 L \longrightarrow (2L)^4. \tag{12}$$

At each step, (i) split the links parallel to the newly refined coordinate, keeping their parent products fixed, and (ii) sample the additional transverse links required at the new sites. Links and plaquettes already represented on the coarser geometry supply conditioning information. The recursive lower-dimensional constructions described in Ref. [3] can be used for the transverse-link generators, although simpler conditional coupling layers could be tried first. In the product-blocking case, subsequent refinements introduce detail links without changing the original W paths.

Let D_k denote the independent group-valued details added at refinement stage k, and $D_{<k} = (D_1, \ldots, D_{k-1})$. Here D_k labels *stages*; their collection $D = (D_1, \ldots, D_4)$ is the aggregate detail variable in Eq. (9). The constrained reconstruction is a bijection at fixed W between these independent coordinates and the fine-link configurations satisfying the product constraints (up to conventional boundary/link-order choices):

$$V = \mathcal{I}_1(W, D_1, \ldots, D_4), \qquad \mathcal{B}_1[V] = W. \tag{13}$$

The splitting map $(W, H) \leftrightarrow (V_A, V_B)$ preserves the product Haar measure, so that no configuration-dependent Jacobian is introduced by this algebraic reconstruction. Flow-layer Jacobians are still required.

4.4 Conditional gauge-equivariant coupling layers

Gauge transformations at the newly created fine sites are independent of those at the coarse sites. In the elementary split,

$$H_\mu(X) \mapsto g(2X + \hat{\mu}) H_\mu(X) g^\dagger(2X + 2\hat{\mu}), \tag{14}$$

while the two-link product $W_\mu(X)$ transforms only at its endpoints. The reference measure for each newly introduced link can be Haar measure; it is invariant under independent left and right multiplication. A learned conditional map must be equivariant under the full local gauge group, not merely under transformations acting on the coarse sites. In particular, feeding gauge matrices into an ordinary neural network does not suffice.

One possible equivariant coupling layer selects active independent detail links H and freezes the other detail links. Using oriented staples S_j built from W, already constructed paths, and frozen details, form an appropriately oriented closed loop and update it through an invertible conjugation-equivariant transformation as in Eq. (5). The updated H' replaces H, and any dependent companion link is rebuilt as $V'_A = W(H')^\dagger$. Thus the hard coarse-link constraint survives *each* coupling layer. Masks and staple constructions must respect the geometric placement and endpoints of the active links. The precise invertible SU(3) layer, including its log-Jacobian, is an architectural choice rather than a consequence of equivariance alone.

4.5 Level-wise likelihood and training procedure

The exact conditional density admits the chain-rule factorization

$$p(D \mid W) = \prod_{k=1}^{4} p(D_k \mid W, D_{<k}). \tag{15}$$

Model each factor by a conditional invertible flow,

$$D_k = F_{1,k}(z_k; W, D_{<k}), \qquad q_\theta(D \mid W) = \prod_{k=1}^{4} q_{\theta_k}(D_k \mid W, D_{<k}), \tag{16}$$

with independent reference variables z_k distributed according to products of normalized Haar measures. Densities and Jacobians refer to these group measures. For a differentiable invertible conditional flow,

$$\log q_{\theta_k}(D_k \mid W, D_{<k}) = \log r_k(z_k) - \log\left|\det J_{F_{1,k}}(z_k; W, D_{<k})\right|. \tag{17}$$

In practice, each paired training example $(W^{(i)}, V^{(i)})$ is decomposed *deterministically* into its target detail sets $D_k^{(i)}$. The loss becomes

$$\boxed{\mathcal{L}_1(\theta) = -\frac{1}{N}\sum_{i=1}^{N}\sum_{k=1}^{4} \log q_{\theta_k}\left(D_k^{(i)} \mid W^{(i)}, D_{<k}^{(i)}\right).} \tag{18}$$

For each level the conditioning details are those of the *training configuration* (teacher forcing), not noisy samples generated by previous levels. This permits separate initialization or training of the four conditional models; end-to-end validation remains necessary because sampling uses generated $D_{<k}$. Importantly, neither the blocked action nor the induced action for V is needed for this supervised objective.

4.6 Sampling, further refinement, and validation

To sample given a fixed input W, draw $z_1, \ldots, z_4$, generate the D_k in increasing order using Eq. (16), and reconstruct V by Eq. (13). An optional final invertible equivariant map $D \mapsto G_\phi(D; W)$ can introduce correlations between detail variables generated at different levels, provided it acts *only* on independent coordinates and all dependent links are rebuilt. An unconstrained flow on the completed fine links would in general destroy $\mathcal{B}_1[V] = W$ and is therefore unsuitable for Flow 1.

There are three separate validation requirements: (a) check $\mathcal{B}_1[V] = W$ to numerical precision on every generated sample; (b) test covariance by gauge transforming inputs and comparing

appropriately coupled outputs or their distributions; and (c) compare observables and conditional statistics of generated V against the native paired ensemble, including short Wilson loops and correlations among newly introduced links. The ability to evaluate $q_\theta(V \mid W)$ is useful but is not itself evidence that the full composed $W \to V \to U$ sampler has the correct long-distance physics.

A practical progression is Haar splitting with simple staple-conditioned layers, followed by the four-level correlated conditional model, and finally the optional fixed-W detail refinement. This retains Abbott *et al.*'s multiscale geometry while making both the training criterion and the hard constraint specific to our paired-data problem.

5 Flow 2: from smoothed links V to original links U

5.1 A distinct conditional problem

Flow 2 is trained using *native original fine configurations and their corresponding smoothed fields*, both on large fine-lattice volumes:

$$U \sim p_{\beta_f}(U) \quad \text{on a large fine lattice,} \qquad V = \mathcal{B}_2[U]. \tag{19}$$

No coarse input W is needed to train this stage, and Flow 2 need not be trained jointly with Flow 1. Its statistical target is

$$p_2(U \mid V) \quad \text{induced by} \quad p_{\beta_f}(U) \text{ and } V = \mathcal{B}_2[U]. \tag{20}$$

The model has the schematic form $U = F_{2,\theta_2}(V, z_2)$, and training on (V, U) pairs can be based on conditional likelihood when a valid density parameterization is available. The underlying forward transformation is assumed fixed; the task is to learn its conditional inverse distribution, including correlations among unsmoothed links.

There is an architectural subtlety worth keeping explicit. If $\mathcal{B}_2$ is a deterministic dimension-preserving smoothing map, the exact conditional distribution $p(U \mid V)$ is supported on the preimage $\mathcal{B}_2^{-1}(V)$, which may be lower-dimensional or even contain only isolated points. A conventional full-dimensional flow density $q_2(U \mid V)$ need not have that support. Thus two implementations should be distinguished:

- **Hard-constrained Flow 2:** construct a parameterization of the preimage of the given $\mathcal{B}_2$, if one is available, and learn a distribution over its free variables. This guarantees $\mathcal{B}_2[U] = V$.
- **Approximate conditional Flow 2:** train a gauge-equivariant generative model on paired (V, U) fields without claiming exact inversion. Quantify the residual $\mathcal{B}_2[U] - V$ and its impact on the composition. Conditional likelihood and reconstruction-type losses may be used only with a clearly defined probabilistic model.

Which option is viable depends on the already selected smoothing/blocking map. Nothing in the definition of Flow 1 requires settling this question now. In particular, we do *not* identify Flow 2 with an unsmearing algorithm based on an assumed analytic inverse.

5.2 Composition and what is preserved

Given a sampled coarse configuration W, the final output is

$$V = F_{1,\theta_1}(W, z_1), \qquad U = F_{2,\theta_2}(V, z_2). \tag{21}$$

Flow 1 guarantees $\mathcal{B}_1[V] = W$ by construction. If Flow 2 also satisfies $\mathcal{B}_2[U] = V$ exactly, then the full relation

$$(\mathcal{B}_1 \circ \mathcal{B}_2)[U] = W \tag{22}$$

necessarily follows. If Flow 2 is approximate, Eq. (22) becomes a validation observable, *not* an identity. This cleanly separates the exact geometry of product splitting from the statistical inverse of the chosen smoothing map.

A second limitation is independent of flow quality: the marginal distribution of coarse Wilson fields generated at β_c can differ from the blocked marginal in the fine training ensemble. Both flows can learn their native conditional distributions accurately and still produce a composed ensemble with a biased coarse marginal. Long-distance matching must therefore be assessed separately.

6 Proposed numerical sequence

The two stages can be built and evaluated independently before composing them.

1. **Fix the forward transformations.** Use the agreed $\mathcal{B}_2 : U \mapsto V$ and product block $\mathcal{B}_1 : V \mapsto W$ throughout. Select matched coarse/fine Wilson couplings and generate independent reference ensembles.

2. **Train Flow 1** on paired (W, V) fields derived from fine configurations as in Eq. (10). Test exact product-block preservation, gauge covariance, conditional likelihood where defined, and observables of smoothed configurations.

3. **Train Flow 2 independently** on large-volume native (V, U) pairs from Eq. (19). Compare generated and reference original configurations conditioned on the same V; explicitly measure blocking consistency $\mathcal{B}_2[U]$ versus V.

4. **Compose the models.** Starting from coarse W drawn at β_c, generate V with Flow 1 and U with Flow 2. Compare final Wilson-loop, plaquette, gradient-flow, and long-distance observables against independently thermalized fine ensembles. Check the full-block residual in Eq. (22).

5. **Measure rethermalization as a diagnostic.** If needed, apply a short fine-action MCMC evolution and track how different observables relax. Compare against deterministic Endres-type interpolation and a simple random-detail Flow 1 baseline. MCMC updating will in general change the original coarse link constraints; this does not invalidate their role in the initial prolongation.

The most informative comparison is whether realistic conditional detail fields reduce the rethermalization cost and preserve long-distance observables better than inexpensive deterministic prolongation at comparable computational cost. Because Flow 2 is trained on large native configurations, its scaling and transfer to the volumes generated by Flow 1 require particular attention.

7 Questions for collaborators

- What gauge-equivariant architecture is expressive and tractable for the Flow 1 detail variables, including correlations among neighboring fine links?
- Given the fixed smoothing transformation $\mathcal{B}_2$, is a hard-constrained second inverse feasible, or should Flow 2 be treated as an approximate conditional generator with a measured consistency residual?

- How well do models trained on large native (V, U) pairs generalize when V is generated by Flow 1 rather than obtained directly by smoothing a native U?
- How accurately must the independently sampled coarse Wilson ensemble match the actual blocked fine ensemble for the resulting long-distance observables to agree?
- What is the end-to-end computational gain relative to Endres-type interpolation followed by fine-lattice rethermalization?

Scope. The two-stage decomposition and the forward blocking maps are taken as design inputs. This note proposes the conditional flow implementation and tests; it makes no claim that both stages already provide exact probabilistic inverses or that rethermalization can be omitted.

References

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